Zilong Li

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E-mail: zilongli@iastate.edu

SPECIALIZATION

Aerospace Engineering with a focus on
Computational Fluid Dynamics , Machine Learning, Reduced Order Modeling, Reinforcement Learning, Fluid-Structure Interactions, Turbulence, and High-Performance Computing.

EDUCATION

Iowa State University (ISU) 08/2021-Present
Ph.D. in Aerospace Engineering Major GPA: 4.0/4.0; Overall GPA: 4.0/4.0

The Pennsylvania State University, University Park (PSU) 08/2017-12/2019
M.S. in Civil Engineering Major GPA: 3.96/4.0; Overall GPA: 3.92/4.0
M.S. minor in Computational Science GPA: 4.0/4.0

Shanghai Jiao Tong University (SJTU) 09/2012-03/2015
M.S. in Fluid Mechanics Major GPA: 87/100; Overall GPA: 85/100

Wuhan University of Science and Technology (WUST) 09/2008-06/2012
B.S. in Engineering Mechanics Major GPA: 89/100; Overall GPA: 87/100 Rank: 3/40

RESEARCH EXPERIENCE

08/2021-Present Department of Aerospace Engineering, ISU
Research Assistant in *Aerospace Engineering*
- Lead a project on Reduced Order Modeling (ROM)
- Implemented and validated reduced order models in OpenFOAM
- Machine Learning enabled turbulence models for unsteady aerodynamic flows
- Experience with Reinforcement Learning

08/2017-12/2019 Department of Civil and Environmental Engineering, PSU
Research Assistant in *Civil Engineering*
- Collaborated on a project on sediment transport
- Implementation of Smoothed Particle Hydrodynamics (SPH) method in DualSPHysics for granular flow
- High-fidelity numerical simulation and analysis of turbulence

06/2015-12/2015 Estuarine and Coastal Science Research Center, Shanghai
Research Assistant in *Computational Fluid Dynamics*
- Lead a field investigation of flow around groins in Hangzhou Bay
- Completed experimental investigation of flow around artificial groins and numerical simulation of the flow field
- Experience with High Performance Computing (HPC)

03/2013-03/2015 Department of Engineering Mechanics**,** SJTU
Research Assistant in *Computational Fluid Dynamics*
- Lead a project on flow characteristics around spur dikes in an open channel

- Rich experience in computer programming, algorithms, and numerical analysis
- Enabled large scale simulations on multiple CPUs via OpenMP

06/2013-03/2014 Ministry of Education (MOE) Key Laboratory of Hydrodynamics, SJTU
Research Assistant in *Particle Image Velocimetry (PIV) Experiment*

- Independently conducted a PIV experiment to measure the flow field
- Assisted in a project on flow measurements around bridge piers

09/2011-06/2012 Department of Engineering Mechanics, WUST
Research Assistant in *Engineering Mechanics*

- Assisted in a project on carbon fiber reinforced concrete design
- Gained the solid knowledge of Finite Element Method (FEM)

PUBPLICATIONS

1. **Z. Li**, L. Fang, A. Sharma, P. He. Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. Accepted by the *AIAA Journal*. Doi: 10.48550/arXiv.2511.18276
2. **Z. Li**, L. Fang, A. Sharma, P. He. Field Inversion Machine Learning for Accurate Predictions of Time Resolved Unsteady Flows Over Airfoils. *AIAA SciTech Forum*, 2025. Doi: 10.2514/6.2025-2049
3. **Z. Li**, P. He. Accelerating Unsteady Aerodynamic Simulations Using Predictive Reduced order Modeling. *Aerospace Science and Technology*, 2023. Doi: 10.1016/j.ast.2023.108412
4. **Z. Li**, P. He. Airfoil Unsteady Aerodynamic Analysis Using a Galerkin Reduced order Modeling Approach. *AIAA SciTech Forum*, 2022. Doi: 10.2514/6.2022-0080
5. **Z. Li**, J. Kou, & J. Zhang. Numerical simulations of flows around a single spur dike in an open channel. *Chinese Journal of Computational Mechanics,* 2016. Doi: 10.7511/jslx201602016

PAPERS IN PREPARATION

- ***Z. Li***, L. Fang, A. Sharma, and P. He. "Augment RANS Turbulence Models for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall Using High Fidelity DES/LES data." (In preparation).

TEACHING EXPERIENCE

01/2022-Present Department of Aerospace Engineering, ISU
Teaching Assistant in *Aerospace Systems Integration, Advanced Flight Structures, Design of Aerospace Systems, Modern Design Methodology with Aerospace Applications*

COMPUTER SKILLS

Programming Languages: Strong proficiency in C/C++, Python, Fortran, MATLAB
CFD Packages: Extensive experience with OpenFOAM
High Performance Computing Platform: TACC Stampede3, NOVA Cluster
Operating Systems: Expertise in Linux (Ubuntu, Fedora)

AWARDS & HONORS

2025 Research Excellence Award in the College of Engineering, ISU
2018 H. Marcus Dean's Chair Scholarship in the College of Engineering, PSU

2018 Harry G. Miller Scholarship in the College of Engineering, PSU
2017 University Graduate Fellowship (UGF) in the College of Engineering, PSU
2014 2nd Prize Scholarship (Top 15%), SJTU
2013 1st Prize in National Graduate Mathematical Modeling Contest (Top 3%)
2012 Outstanding Graduate Student Award (Top 10%), WUST
2011 3rd Prize in National Undergraduate Mathematical Contest in Modeling (Twice)
2010 National Encouragement Scholarship (Top 5%), WUST
2010 3rd Prize in National College Student Basic Mechanics Experiment Contest (Top 3%)
2010 "Excellent" Prize in National Pei-Yuan Chou Mechanics Competition

EXTRACURRICULAR ACTIVITIES

2013-2014 **Assistant**, Employment Guidance Center, SJTU
2010-2012 **Vice-Chairman**, Mathematical Modeling Association, WUST
2009-2012 **Core Member**, Wuhan University of Science and Technology Table Tennis Club

REFERENCES

Dr. Ping He
Assistant Professor, Department of Aerospace Engineering,
Iowa State University, Ames
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Dr. Anupam Sharma
Professor, Department of Aerospace Engineering,
Iowa State University, Ames
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